

Practice Problems

Practice Problems: WACC

- **Question 1:** Ignoring the tax deductibility of interest, a company has \$8m of equity and \$4m of corporate debt. The expected returns are 6% for its debt and 12% for its equity.
 - What is the expected return on its operations?
- **Question 2:** A company has \$12B of equity and \$6B of debt. Its debt and equity have expected returns of 4% and 15%, respectively. The corporate tax rate is 25%.
 - What is the expected return on its operations?

WACC Cookbook

Discount rate (WACC) cookbook

To determine the discount rate for an actual project, investment, or company:

- Find (one or more) publicly-traded company with similar risk profile (“comparable company”)
 - A publicly-traded company can be its own comparable company
- Use beta from comparable to calculate its expected return to equity (r_E), using CAPM
- Determine yield on comparable’s debt (r_D), typically from similarly rated corporate bonds
- Find market value of comparable’s equity (E) (cannot use book value from balance sheet)
- Find value of comparable’s debt (D) (*can* use book value from balance sheet)
- Use WACC equation to get discount rate (*i.e.*, expected return on comparable’s assets)

Amazon (AMZN)'s WACC

- Use CAPM to get AMZN expected return on equity (r_E):
 - Amazon has an equity beta of 1.45
 - Yield on ten-year US bonds as risk-free rate ($r_f = 4.47\%$)
 - $r_E = 4.47\% + 1.45 \times 6.0\% = 13.17\%$
- Use bond yield for expected return on debt (r_D):
 - Amazon's bonds are rated AA and are considered safe
 - Current yield on AA-rated corporate bonds is 5.07%
- Value of Amazon's equity (E) is \$2.58T (market cap.)
- Value of Amazon's debt (D) is \$130.9B





Amazon.com Inc

✓ Added

\$239.20 ↑ +42.94% (+71.86) 5Y

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Overview Earnings Financials Holdings New

Amazon declined approximately 8% over the past month, pressured by a massive \$200 billion capital exp...

Open	\$243.61	Volume	16.71M	Beta	1.45
High	\$250.43	P/E ratio	28.61	Shares outstanding	10.76B
Low	\$237.00	52-wk high	\$278.56	No. of employees	2M
Mkt. cap	2.58T	52-wk low	\$196.00		
Avg. vol.	44.07M	EPS	\$8.37		

Risk-free rate (r_f)

- CAPM calculation requires risk-free rate (r_f)
- We typically use yield on 10-year US government bond:
 - <https://www.bloomberg.com/markets/rates-bonds/government-bonds/us>

Treasury Yields

Name ▼	Coupon ▼	Price ▼	Yield ▼
⊕ GB3:GOV 3 Month	0.00	3.62	+3.70%
⊕ GB6:GOV 6 Month	0.00	3.67	+3.79%
⊕ GB12:GOV 12 Month	0.00	3.68	+3.83%
⊕ GT2:GOV 2 Year	4.00	99.89	+4.06%
⊕ GT5:GOV 5 Year	4.13	99.75	+4.18%
⊕ GT10:GOV 10 Year	4.38	99.27	+4.47%
⊕ GT30:GOV 30 Year	5.00	100.50	+4.97%

Corporate bond yields (2026-06-10)

	CLOSE	% CHG	YTD TOTAL RETURN	52-WK % CHG	YIELD (%), 52-WEEK RANGE		
					LATEST	LOW	HIGH
Broad Market Bloomberg Fixed Income Indices							
U.S. Government/Credit	2609.31	0.23	-0.13	4.47	4.690	4.090	4.760
U.S. Aggregate	2248.52	0.23	-0.01	5.01	4.780	4.160	4.970
U.S. Corporate Indexes Bloomberg Fixed Income Indices							
U.S. Corporate	3452.70	0.24	0.21	5.65	5.240	4.680	5.340
Intermediate	3388.08	0.16	0.24	5.14	4.960	4.330	5.020
Long-term	4456.68	0.43	0.16	6.77	5.860	5.370	6.020
Double-A-rated (AA)	647.52	0.25	-0.23	4.50	5.070	4.410	5.170
Triple-B-rated (Baa)	949.74	0.24	0.48	5.99	5.400	4.860	5.510
High Yield Bonds ICE Data Services							
High Yield Constrained*	625.43	0.02	1.32	6.64	7.198	6.489	7.668
Triple-C-rated (CCC)	629.93	0.09	0.25	5.29	13.245	11.438	14.020
High Yield 100	4151.46	0.02	0.77	5.19	6.504	5.828	7.136
Europe High Yield Constrained	400.07	0.09	1.23	3.57	5.449	4.820	6.198
Global High Yield Constrained	545.94	0.04	1.72	6.61	6.908	6.317	7.438

- <https://www.wsj.com/market-data/bonds/benchmarks>

Debt (D): Amazon (AMZN) balance sheet

- Yahoo finance states Amazon's total debt of \$209.888B

Amazon.com, Inc. (AMZN) ☆				
246.46 +7.91 (+3.32%)				
As of 1:54:05 PM EDT. Market Open.				
Income Statement Balance Sheet Cash Flow Dividends View in AlphaSpace				
All numbers in thousands				
Annual Quarterly				
Breakdown	3/31/2026	12/31/2025	9/30/2025	6/30/2025
▼ Total Assets	916,630,000	818,042,000	727,921,000	682,170,000
▼ Total Liabilities Net Minority I...	474,716,000	406,977,000	358,290,000	348,395,000
▼ Total Equity Gross Minority In...	441,914,000	411,065,000	369,631,000	333,775,000
Total Capitalization	560,988,000	476,713,000	420,373,000	384,493,000
Common Stock Equity	441,914,000	411,065,000	369,631,000	333,775,000
Capital Lease Obligations	90,814,000	87,339,000	84,677,000	83,221,000
Net Tangible Assets	418,465,000	378,595,000	346,371,000	310,620,000
Working Capital	38,399,000	11,078,000	1,670,000	4,499,000
Invested Capital	560,988,000	476,713,000	420,373,000	384,493,000
Tangible Book Value	418,465,000	378,595,000	346,371,000	310,620,000
Total Debt	209,888,000	152,987,000	135,419,000	133,939,000
Net Debt	17,258,000	--	--	--
Share Issued	11,269,251.80	11,246,000	11,202,000	11,175,000
Ordinary Shares Number	10,754,251.80	10,731,000	10,687,000	10,660,000
Treasury Shares Number	515,000	515,000	515,000	515,000

Debt (D): Amazon (AMZN) balance sheet

Amazon's quarterly filing (1Q2026) shows:

- \$119.074B of long-term debt
- \$90.814B of long-term lease liabilities
 - Long-term lease liabilities are a kind of debt
- Total debt is \$209.888B

	December 31, 2025	March 31, 2026
ASSETS		
Current assets:		
Cash and cash equivalents	\$ 86,810	\$ 101,816
Marketable securities	36,219	41,273
Inventories	38,325	36,534
Accounts receivable, net and other	67,729	75,532
Total current assets	229,083	255,155
Property and equipment, net	357,025	397,458
Operating leases	86,054	88,741
Goodwill	23,273	23,449
Other assets	122,607	151,827
Total assets	\$ 818,042	\$ 916,630
LIABILITIES AND STOCKHOLDERS' EQUITY		
Current liabilities:		
Accounts payable	\$ 121,909	\$ 124,749
Accrued expenses and other	75,520	71,120
Unearned revenue	20,576	20,887
Total current liabilities	218,005	216,756
Long-term lease liabilities	87,339	90,814
Long-term debt	65,648	119,074
Other long-term liabilities	35,985	48,072

Amazon (AMZN) WACC

$$WACC = \frac{D}{D + E} (1 - t_c) r_D + \frac{E}{D + E} r_E$$

$$= \left(\frac{\$210B}{\$210B + \$2.58T} \right) (1 - 25\%) 5.070\% + \left(\frac{\$2.633T}{\$210B + \$2.58T} \right) 13.17\% = 12.47\%$$

- **Note:** $D/(D+E) = 7.5\%$, and $E/(D+E) = 92.5\%$

Practice Problems

Practice Problems: WACC

- **Question 1:** NVIDIA (NVDA)'s equity has a beta of 2.12. What is the expected return on its equity (r_E)?
- **Question 2:** Zoom Communications (ZM)'s equity has beta of 0.69. What is the expected return on its equity (r_E)?
- **Question 3:** Cinemark Holdings (CNK) is a publicly-traded company that operates movie theaters. It's financials are: Equity beta 1.94; market cap. of equity (E) \$3.7B; debt (D) of \$3.5B
 - What is Cinemark Holding (CNK)'s WACC?
- **Assumptions:** Corporate tax rate is 25%. Risk-Free Rate is 4.5%. Market Risk Premium is 6%. Corporate bond yield (r_D) is 5.1% (Bloomberg's average corporate bond index)

DCF/NPV in Practice: Terminal Value (TV)

Terminal Value (TV)

- When valuing a project or business that continue indefinitely, we need to end projections eventually
 - Typical horizon for projections (forecasts) is 5 years
- Conceptually, want to extend projections until project or business is in a stable steady state, i.e., on a constant growth trajectory
 - Alternatively, until there is an exit event, e.g., an IPO or sale for a start-up
- Then, use terminal value (TV) to capture value of future cash flows after this horizon
 - TV often determined using growing perpetuity calculation

Example: Calculating TV

Item (figures in \$'000)		Year 0	Year 1	Year 2	Year 3	Year 4
Revenue (REV)		\$0.0	\$400.0	\$440.0	\$480.0	...
Cost of Goods Sold (COGS)		\$0.0	\$220.0	\$240.0	\$260.0	...
Gross Profit (REV-COGS)		\$0.0	\$180.0	\$200.0	\$220.0	...
Operating Expenses (SG&A)		\$0.0	\$120.0	\$120.0	\$120.0	...
Depreciation (DEPR)		\$0.0	\$40.0	\$40.0	\$40.0	...
EBIT (Gross Profit-SG&A-DEPR)		\$0.0	\$20.0	\$40.0	\$60.0	...
Capital Expenditure (CAPX)		\$120.0	\$0.0	\$0.0	\$0.0	...
dNet Working Capital (dNWC)		\$15.0	\$5.0	\$5.0	\$5.0	...
Corporate Tax Rate (T)	25%					
Free Cash Flow (FCF)		-\$135.0	\$50.0	\$65.0	\$80.0	...

- Similar to example with income statement for calculating FCF, but now project continues indefinitely

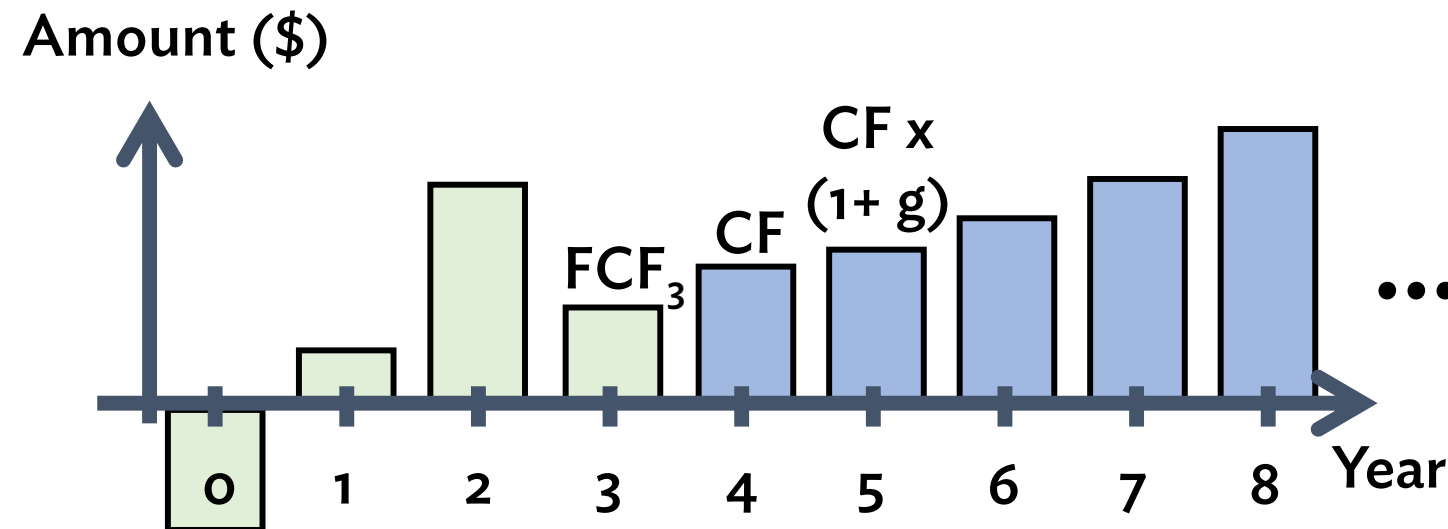
Indefinite project

- The project (or investment) will be stable after three years
 - Our projections (forecast) have a three-year horizon
- After three years, FCFs will grow at the constant rate of $g = 3\%$
- Our valuation captures these future cash flows, after the horizon of our projections, as a growing perpetuity

TV from growing perpetuity

- Green cash flows are included in projections, and blue cash flows are beyond the horizon
- Terminal value (TV) is the present value (PV) (in year 3) of the blue cash flows, calculated using the growing perpetuity formula

$$PV_3 = \frac{CF_4}{r - g}$$



$$TV_3 = \frac{FCF_3(1 + g)}{WACC - g}$$

Back to example: Calculate TV

	A	B	C	D	E	F	G	H	I	J
54										
55				0	1	2	3	4		
56		Item (figures in \$'000)		Year 0	Year 1	Year 2	Year 3	Year 4		
57		Free Cash Flow (FCF)		-\$135.0	\$50.0	\$65.0	\$80.0	...		
58		PV of FCF		-\$135.0	\$44.6	\$51.8	\$56.9	=G57/(1+\$C63)^G55		
59		PV		\$18.4						
60										
61		Terminal Value (TV)					\$915.56	=G57*(1+C62)/(C63-C62)		
62		growth rate (g)	3%							
63		WACC	12%							
64		PV of TV		\$651.67	=G61/(1+C63)^G55					
65										
66		Total PV (FCF + TV)		\$670.08	=D64+D59					
67										
68										

- Calculate TV assuming a growth rate of $g = 3\%$ and a WACC of 12%

Terminal value (TV) comments

- Careful with timing
 - TV is PV of future FCFs as of last period of projections (T)
 - First omitted FCF is in the following period ($T+1$): $FCF_{T+1} = FCF_T \times (1+g)$
 - FCF_{T+1} is the cash flow in the numerator of the growing perpetuity formula
 - Discount TV back to today using the WACC rate, $PV(TV) = TV / (1+WACC)^T$
- Often when valuing companies and projects, most of value is in TV (as in example)
 - Can be mitigated by extending horizon
 - But highlights importance of future value, i.e., g and $WACC$
- More elaborate projection models sometimes have multiple stages of terminal values (TVs)
 - Typically, an interim stage with faster growth followed by a final stage with lower perpetual growth

Practice problems: TV

- **Question 1:** You are evaluating a project with an indefinite life, and you have projected the FCFs for three years. What is the TV? What is the total PV?

- **Assumption:** WACC is 13% and growth rate is 3%

(Figures in \$'000)	0	1	2	3
Free Cash Flow (FCF)	\$0.0	\$50.0	\$70.0	\$100.0

- **Question 2:** When using the growing perpetuity to calculate a TV, the TV is a multiple times the final FCF. When WACC equals 13% and the growth rate is 3%, what is this multiple?

Valuing Companies and Their Equity

DCF valuation of operating assets (OA)

- Valuing operating assets is just like valuing a project
 - Project (forecast) financials in income statement
 - Calculate free cash flows (FCFs)
 - Determine WACC
 - Find the terminal value (TV)
- Discount and sum to get present value (PV)
 - This is present value (PV) of company's operating assets (OA), i.e., the enterprise value (EV)

Balance sheet and company value

- Total company value comprises operating assets (OA) and excess cash (EC)
 - Value of operating assets (OA) is called enterprise value (EV)
 - Excess cash is cash that is free to pay out to investors
- DCF valuation of company's FCFs gives the value of the operating assets, i.e., determines enterprise value (EV)
- Value of total company: $\text{Total Value} = \text{Excess Cash} + \text{Enterprise Value} = \text{Debt} + \text{Equity}$
- Value of equity: $\text{Equity} = \text{Total Value} - \text{Debt}$
 - Often convenient to define $\text{Net Debt} = \text{Debt} - \text{Excess Cash}$
 - Then, $\text{Equity} = \text{Enterprise Value} - \text{Net Debt}$

Assets	Liabilities
Excess Cash	Debt
Operating Assets	Equity

Valuing a company share

- Valuation relation can be rearranged to value a company's equity (E)
 - **Equity = Total Value – Debt**
 - **Equity = Enterprise Value - Net Debt**
- To value an individual share, divide equity valuation by number of outstanding shares
 - If your valuation of a share exceeds the current market price, shares may be undervalued, and you should consider buying some!

Assets	Liabilities
Excess Cash	Debt
Operating Assets	Equity

Outstanding shares

- A company's shares are owned by its shareholders
 - A company can issue new shares when it needs capital (primary market)
 - For publicly-traded companies, shares are bought and sold in the stock market (secondary market)
- **“Authorized Shares”** is the maximum number of shares allowed (by corporate charter)
 - Mostly a theoretical number, not relevant for us
- **“Issued Shares”** are shares that the company has created
 - Companies can issue new shares to sell to investors to raise money for investments and projects
- **“Treasury Shares”** are shares that have been issued but are still owned by the company
- **“Outstanding Shares”** are shares that have been sold and are owned by shareholders
 - $\text{Outstanding Shares} = \text{Issued Shares} - \text{Treasury Shares}$
- To value a share of a company, divide total equity value by number of **outstanding shares**



Amazon.com Inc

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Avg. vol.	44.07M	EPS	\$8.37		

Amazon

- Balance Sheet from 1Q2026 Amazon Quarterly filing

AMAZON.COM, INC.
Consolidated Balance Sheets
(in millions, except per share data)
(unaudited)

	December 31, 2025	March 31, 2026
ASSETS		
Current assets:		
Cash and cash equivalents	\$ 86,810	\$ 101,816
Marketable securities	36,219	41,273
Inventories	38,325	36,534
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Total assets	<u>\$ 818,042</u>	<u>\$ 916,630</u>
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Current liabilities:		
Accounts payable	\$ 121,909	\$ 124,749
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Total current liabilities	218,005	216,756
Long-term lease liabilities	87,339	90,814
Long-term debt	65,648	119,074
Other long-term liabilities	35,985	48,072
Commitments and contingencies		
Stockholders' equity:		
Preferred stock (\$0.01 par value; 500 shares authorized; no shares issued or outstanding)	—	—
Common stock (\$0.01 par value; 100,000 shares authorized; 11,246 and 11,269 shares issued; 10,731 and 10,754 shares outstanding)	112	113
Treasury stock, at cost	(7,837)	(7,837)
Additional paid-in capital	140,024	143,979
Accumulated other comprehensive income (loss)	28,230	24,868
Retained earnings	250,536	280,791
Total stockholders' equity	411,065	441,914
Total liabilities and stockholders' equity	<u>\$ 818,042</u>	<u>\$ 916,630</u>

Amazon shares

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- Amazon (AMZN) has issued 11.269B shares. Of these, 10.754B are outstanding (numbers are in millions)
 - Book value of common shares and total stockholders' equity are unrelated to market cap

Homework Problem

- Your company considers investing in a project with these two-year projections. What are the project's FCFs?
- Your (publicly-traded) company has an (equity) beta of 1.5, it has 40% debt and 60% equity, what is your company's expected return to equity? What is the WACC?
- Assuming the project's terminal growth rate (g) after year 2 is 3%, what is the terminal value (TV)?
- What is the NPV of the project? Should you invest?
- **Assumptions:** risk-free rate of 4%, market risk premium of 7%, corporate debt pays 6%, corporate tax rate of 25%

(Figures in \$'000)		Year 0	Year 1	Year 2
Revenue (REV)		\$0.0	\$45.0	\$75.0
Cost of Goods Sold (COGS)		\$0.0	\$10.0	\$20.0
Gross Profit (REV-COGS)		\$0.0	\$35.0	\$55.0
Operating Expenses (SG&A)		\$0.0	\$5.0	\$5.0
Depreciation (DEPR)		\$0.0	\$10.0	\$10.0
EBIT		\$0.0	\$20.0	\$40.0
Capital Expenditure (CAPX)		\$80.0	\$0.0	\$0.0
dNet Working Capital (dNWC)		\$5.0	\$5.0	\$5.0